

Machine Learning and Data Mining - Final Project

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1. Introduction

The purpose of this project is to apply various data mining and machine learning algorithms to a selected dataset. Specifically, the results of a survey of Spanish graduates from the 2013/2014 academic year, collected in 2019 by the National Institute of Statistics, will be studied.

The aim is to investigate whether there is any relationship between their sociodemographic profile, qualifications obtained, linguistic or digital skills, different training activities and their current employment and personal situation, including difficulties in finding employment, average salary, mobility or domestic situation.

In this way, the data will be used to try to predict the situations that those currently studying in Spain might encounter in the future.

All figures and files generated during the execution of the project can be located in the corresponding [OneDrive folder](#).

2. Descriptive Statistics

The dataset can be found in [Kaggle.com](#). It compiles 298 attributes, the vast majority of them categorical, about 31,651 Spanish graduates surveyed. The attributes are classified into 7 blocks:

TO Personal and sociodemographic data

BE Education and learning of the university graduate

This is the largest section. It contains information about:

a) Scholarships

b) University studies *c)*

Vocational training cycles *d)*

Languages

and Courses

CM Mobility

DC Current situation of university graduates

AND University graduate's work history

FH Home Information

GA Additional variables

In many cases, a high proportion of NA (null) values are found, mainly because there are dependencies between attributes. For example, if a respondent answers "no" to whether they received any scholarships, all related attributes will be affected.

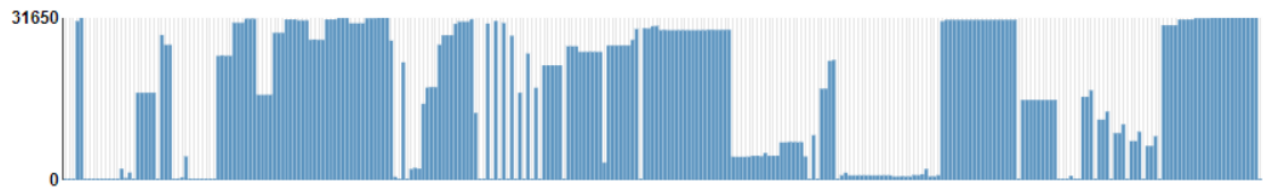


Figure 1: Proportion of null values per attribute

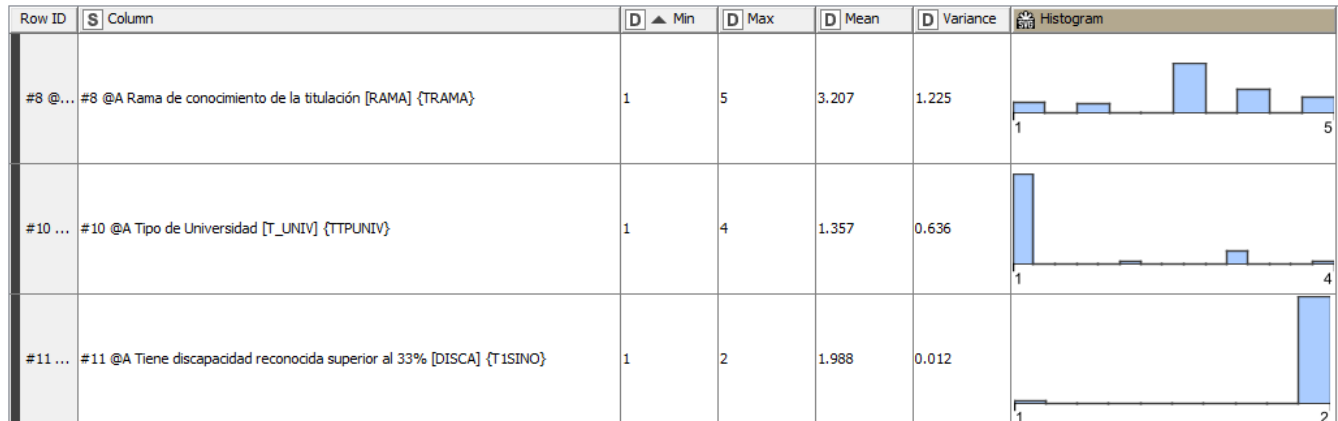


Figure 2: Frequency of the different classes in some attributes

The types of scholarships received will not be applicable. Figure 1 shows the number of null values for each attribute.

Furthermore, in certain attributes lacking values of this type, the classes exhibit a high skew in their values. Figure 2 shows some of the attributes with this characteristic, along with others that are better distributed.

Figures 3 and 4 also show a visualization of the first 2 PCA components, as well as a two-dimensional projection generated by the t-SNE method, using the study branch to label the colors.

Finally, it is interesting to observe the covariance matrix of all the attributes in the dataset, shown in Figure 5, where the different attribute blocks have been delimited. This also reveals that the correlation between attributes in different blocks is, in general, quite weak, which will make the classification task difficult.

3. Preprocessing

Some of the data mining applications that have been used require data in normalized numeric format, while others need unique classes.

Furthermore, most attributes are coded, which creates complications both in understanding the values and in applying numerical techniques when the data are arbitrarily separated by the assignment of codes into subcategories, such as those used by attribute 7, of the Scope of study, which represents the data using 4 digits in a hierarchical way, sometimes omitting the last one if it is 0.

This method of assigning codes creates an artificially skewed spatial distribution in the data, and therefore it has been necessary to adapt the numerical transformation accordingly. There are also attributes with non-numeric codes, in particular the CNAE attribute for the classification of economic activities. In this case, a unique index is assigned to each alphanumeric code.

On the other hand, the large number of attributes and their coded names make it very difficult to work with them or generate graphs and association rules.

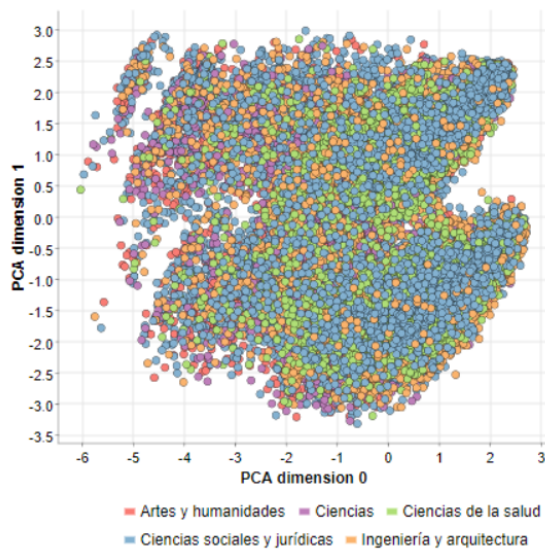


Figure 3: First 2 dimensions of PCA

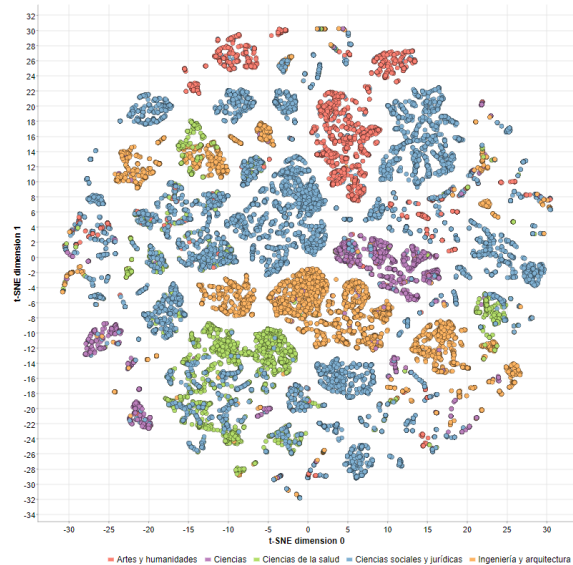


Figure 4: t-SNE of the 2-dimensional dataset

For these reasons, an initial data preprocessing phase was performed, in which all available information was extracted from the legend and condensed into attribute headers. The generated headers use the format #id @BLOCK Description [CODE]{GUY} This allows access to all these attributes as needed. In particular, this format makes it easy to filter by attribute blocks or types.

Additionally, a second table has been created from the legend, containing the translation of the codes for each data type, as well as a component that allows replacing the codes using this code dictionary. This table has also been used to normalize all codes associated with NA and NS/NC values as null values, since the code denoting this value is different for each data type and is often an extreme numeric value. For some applications requiring numeric data, null values have been normalized as -1, because the large proportion of null elements makes it difficult to ignore them without losing a large portion of the information.

4. Models used

The only numerical attributes that could be used for regression are not considered particularly relevant. On the other hand, to apply the knowledge acquired, a pseudo-regression could be performed on salary or the number of years until finding a job, replacing each category with its numerical mean or similar value.

Therefore, the main tasks to be performed are classification, clustering, and association rule finding. Specifically, the aim is to experiment using the following algorithms:

- Clustering
 - k-Means
 - Expectation-Maximization
- Association rules
- Classification
 - Logistic regression
 - SVM
 - Neural networks

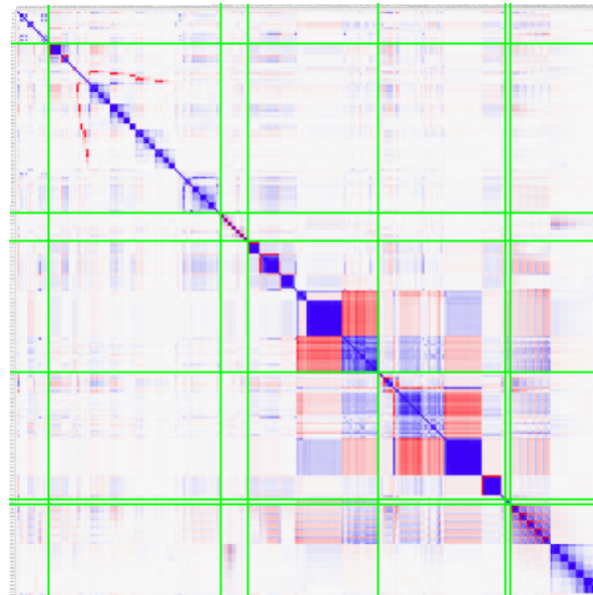


Figure 5: Covariance matrix

- RProp MLP Learner
- Keras neural network
- k-Nearest Neighbours
- Random forest

It will be seen later that not all of them are applicable due to the dimensions of the dataset, although they are applicable due to their format.

5. Development

5.1. Clustering

The process begins by applying an unsupervised model to gain an overview of the data and the main groupings, before proceeding with supervised models for further analysis. For this exercise, all data is used without partitioning into train and test groups, as the objective is visualization, not prediction.

The first difficulties that arise when applying segmentation to the dataset are the large number of attributes, which complicates the interpretation of the results as well as obtaining representative clusters, and the selection of a k appropriate.

To remedy the first problem, the idea of applying PCA arises. However, if this is done, there is no way to know which actual attributes influence the clustering. Therefore, certain sets of attributes are manually filtered to obtain more meaningful clusters in regions of interest, as an extra step in addition to segmenting the entire dataset.

Regarding the second problem: Weka's EM node allows selection k dynamically, but those implementing k-Means do not. For this reason, the elbow method is applied to k-Means using a loop that obtains, for a range of values of k , the sum of the squared Euclidean distances from each point to the centroid of its cluster. The results are plotted on a graph to select the k more visually appealing. The implementation is inspired by the one found on Knime Hub.

One problem identified is that the algorithm sometimes fails to progress beyond a local optimum, causing the graph to exhibit peaks instead of a downward slope. When this occurs, the number of iterations is increased and/or a different random seed is used until the result is reasonable.

Once the clustering is performed, the results are interpreted graphically by representing the values corresponding to the centroids in heat maps that allow us to appreciate in which attributes there is more contrast between clusters.

5.1.1. k-Means

Global analysis

The choice is made $k=7$, although there is no clear elbow in the resulting graph after applying the elbow method.

The heat map representing the centroids is quite large due to the high number of attributes, but fortunately, the aim is not to conduct an exhaustive analysis but rather to identify the main differences between clusters. These differences appear in age group, whether they pursued other studies (another Bachelor's/Diploma/Bachelor's or Master's degree), whether they are unemployed, whether they had any paid work during their studies, and in data related to their affiliation and contributions to Social Security between 2015 and 2019.

Unemployed individuals not affiliated with Social Security are distinguished from those who are affiliated. Differences also exist between the two groups regarding whether they had paid employment upon completion of (or during) their studies and job search.

However, these results vary from run to run depending on the random seed chosen and are therefore only indicative of which attributes allow the sample to be separated into groups, and which do not seem to be so relevant.

Filtered

Now, the dataset is filtered to see the resulting clusters, considering only the degrees and current employment status of the respondents, adding information about scholarships received. Applying the elbow method again, the following is chosen $k=6$.

Figure 6 shows the normalized values of the initial categorical attributes. To interpret their meaning, consider the original dataset legend and its translation: values closer to 0 indicate that the attribute is not applicable. In yes/no questions, a low value represents "yes," a high value "no," and there are no "NA" responses. In questions answered with an approximate number, such as the number of other studies completed, lower values represent lower ranges and vice versa, and there are "NA" responses for those who have not completed other studies. The value close to 1 for "Current work schedule type" is due to a large proportion of "NS/NC" values.

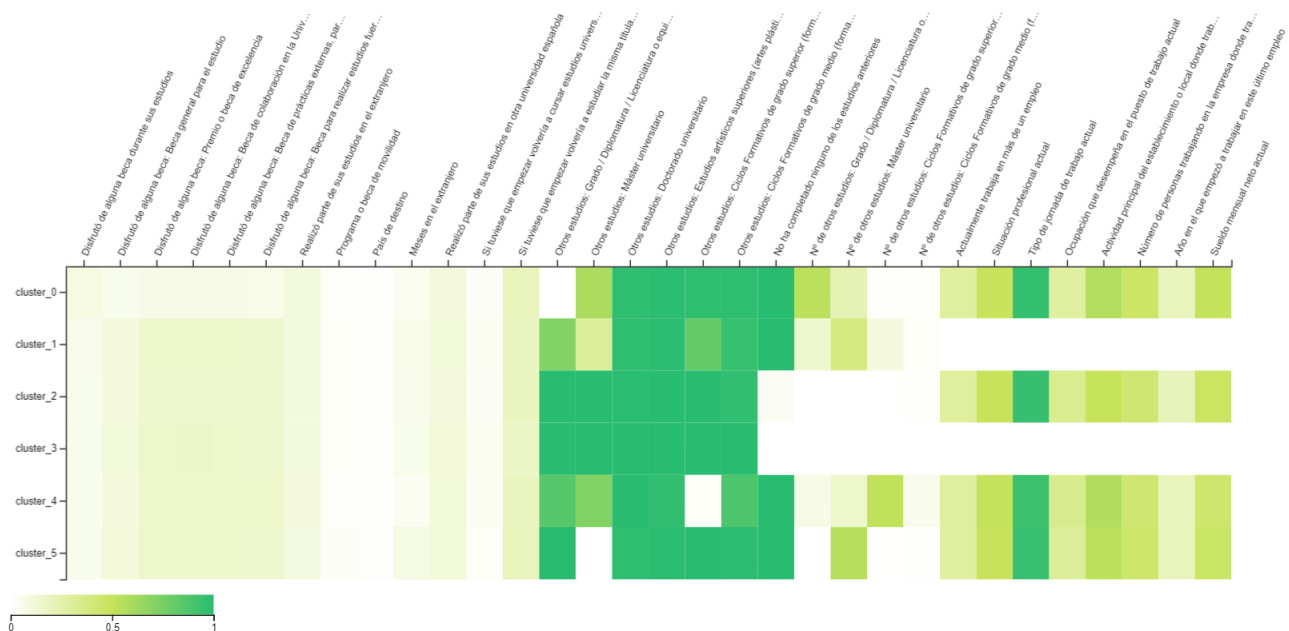


Figure 6: Clustering on attributes of blocks B and D

The data have been segregated mainly according to the number of other studies (Bachelor's, Diploma, Licentiate or

(equivalent), if the graduate completed a master's degree and partially if they completed advanced vocational training cycles in vocational training, plastic arts and design, or advanced sports education. Employed individuals were also separated from unemployed individuals (employment status attributes left blank, corresponding to the value NA), as were those who had completed their studies from those who had not, with differences in the number of other studies.

No significant differences were observed between clusters regarding scholarships awarded or current employment status, other than whether the graduate was currently employed or not. After repeating the experiment with different random seeds, the results were similar. In other words, scholarships do not appear to have a strong impact on employment status. On the other hand, there is a subtle difference in scholarship attributes among those who completed more than one degree: it seems that, in general, a greater number of them received at least one scholarship.

In summary, the main trends observed in each cluster are indicated, along with their relative size:

- Cluster 5 (29.92%): graduates with a Master's degree, but without further qualifications.
- Cluster 2 (25.762%): graduates without further studies, employees.
- Cluster 0 (19.137%): mainly graduates with more than one Bachelor's degree, some with a Master's degree, and without further studies.
- Cluster 4 (10.878%): with higher-level vocational training cycles.
- Cluster 1 (9.712%): unemployed/inactive people with more studies of different types.
- Cluster 3 (4.591%): graduates without further studies, unemployed/inactive.

This does not mean that each cluster exclusively includes these groups, but rather that these are the attributes that stand out most in each of them.

5.1.2. Expectation-Maximization

EM cannot be applied to this dataset due to its high time complexity: Weka's EM node only completes execution in a reasonable time when 5% or less of the data is used. Specifically, 5% of the data filtering only block E (62 attributes). This is a small sample size and therefore not worth performing the analysis with this algorithm.

Even so, the possibility offered by the node to select was explored. *kdynamically by crossvalidation*, comparing it with the implementation of the elbow method by selecting *akstatic* in each iteration. The version for k-Means could be directly adapted simply by adding a step to calculate the centroids and making a few other adjustments.

5.2. Association Rules

To find association rules, two nodes were tested: Knime's Association Rule Learner and Weka's Predictive Apriori. After comparing the output format of both, Association Rule Learner was ultimately chosen to allow manipulation of the results and filtering of relevant rules according to specific criteria. Another important reason for this choice is that Predictive Apriori cannot handle the data types of all columns in the dataset, even after preprocessing.

The node output only displays attribute values, so to interpret the resulting rules, a prefix must be added indicating the attribute corresponding to each cell's value, since many of the 298 attributes share the same categories. This adds to the functionality of the translation component.

A support threshold is chosen $\tau = 0,5$ and trustworthy $\gamma = 0,9$. Ideally, the support threshold would be the lowest possible, to obtain rules that also apply to minority subsets, such as the unemployed, but below 0,5 The execution time is no longer acceptable.

Furthermore, to prevent the predominant attributes in the dataset from always appearing in the rules, those with a variance less than 0.05 are filtered out before applying the algorithm. Finally, for better visualization of the results, a component is added that allows filtering the resulting rules by the consequent block and the premises.

Despite the measures applied, most of the resulting rules involve prevailing values and, consequently, do not provide much information. For example, "Country of residence: Spain $\Rightarrow$ "Working" was used when the aim was to find relationships between current employment status and the graduate's field of study or additional education. With greater computing power, the support threshold could have been lowered enough to obtain some rules related to these attributes.

The following are examples of some of the rules obtained after filtering the dataset. Without filtering, the most common rules are similar to those shown here. No rules with consequents were obtained in blocks B, C, or F.

Filtering for consequent in D and premises in A, B and E did not find premises in B, only in A and E:

Premises	Consequent	τ	γ
E. Has been working for 2 or more years	Working	0,779	0,918
E. Has been working for 2 or more years E. Did not seek employment through other means E. Did not seek employment through temporary employment agencies	Working	0,722	0,922

With consequents in A and E and premises in D, that is, inverting the blocks of the consequent and the premises:

Premises	Consequent	τ	γ
Working	E. Has been working for 2 or more years	0,779	0,909
Working Working in Spain	A. Lives in Spain	0,777	0,966

5.3. Classification

Given the large number of attributes in the dataset, there are a high number of possible classification tasks. However, by its very nature, for the purpose of classification, we are only truly interested in situations where we predict an attribute of blocks D or E based on information from blocks that can be known a priori, i.e., A and B.

Specifically, 3 attributes have been classified based on their relevance:

- Current net monthly salary
- Time elapsed from the completion of these studies until the start of employment
- Professional situation of your first job

The classification has been abstracted using a metanode that tests each of the 5 methods used and compares their results for each attribute.

In this node, there is an initial preprocessing section that is shared by the different methods, as well as a final section that collects the metrics of each method.

5.3.1. Difficulties

During the development process of the different classification methods, some difficulties have been encountered, mostly as a result of the large number of attributes and rows handled.

For example, the algorithm of logistic regression It is not able to converge unless the dimensionality is reduced beforehand by means of PCA, as has been done preserving 95% of the initial variance, which corresponds to 89 components out of the 118 belonging to blocks A and B.

On the other hand, the number of instances m It is very large in relation to the number of attributes p (31651 with respect to 298), so it is not feasible to apply SVM on this dataset. Even if only 1% of the data is selected, training is expensive even with the polynomial kernel (the simplest available in the Knime SVM Learner node), besides which very poor prediction results would be obtained as it is not a representative portion.

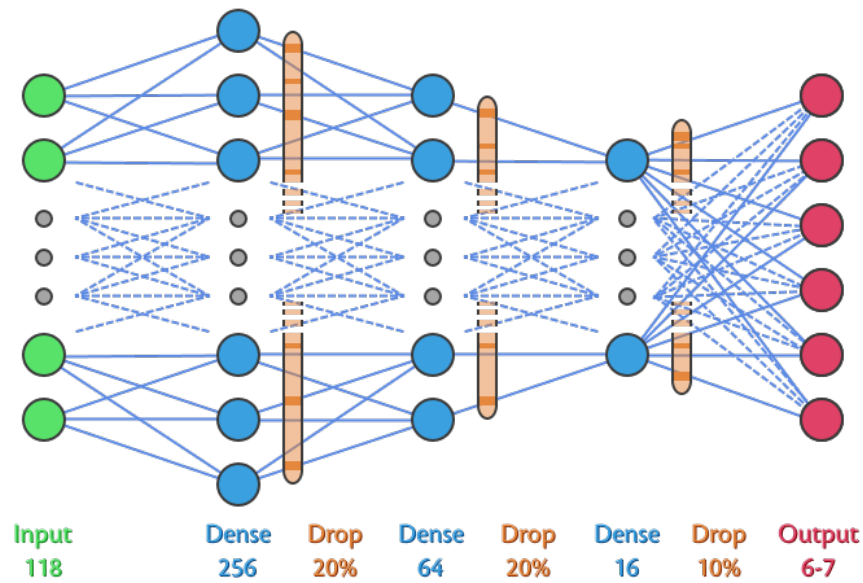


Figure 7: Representation of the Keras network used

Likewise, the large number of output classes for the chosen attributes can sometimes result in classifiers deciding to ignore the less frequent classes, even when taking the stratified data split over the classified attribute.

To address this problem, tests were conducted using the SMOTE technique, which generates synthetic training data to compensate for the low frequency of minority classes using linear interpolation. k neighbors. However, while this technique improved the overall accuracy of the classifiers by performing validation on augmented data, using only the original data for validation reduced accuracy, even when taking $k=1$, that is, only by duplicating data from minority classes. For this reason, the SMOTE technique has not been used in the final models.

Another problem is that in the case of the predictorMLPWith Knime, it is not possible to efficiently obtain a learning curve to evaluate training. Therefore, a classifier was also used. Keras. Thanks to the integration of Knime with Keras, this classifier allows you to determine at what point the network begins to overfit.

The Keras network used consists of 3 dense inner layers with decreasing numbers of neurons: 256, 64 and 16, as shown in Figure 7, using interposed dropout layers of 20%, 20% and 10% respectively, to mitigate or slow down overfitting that may occur during training.

To use the Keras classifier, it was necessary to adapt the class column into multiple binary classification columns, as well as undo this change after the prediction in order to compare the accuracy metrics with the other classifiers.

5.3.2. Parameter Optimization

Parameter optimization strategies are different for each classifier.

In the case of the logistic regression, The main concern is to get the model to converge, which was achieved thanks to dimensionality reduction.

In the case of the classifier by k-Nearest Neighbours, The optimal number of neighbors was determined using cross-validation partitioning. As shown in Figure 8, accuracy increases with the number of neighbors until it converges to a certain value. Therefore, a scaling strategy was used to find the optimal number of neighbors.

In the case of the predictorMLP, Given the large number of attributes, and due to the training time of

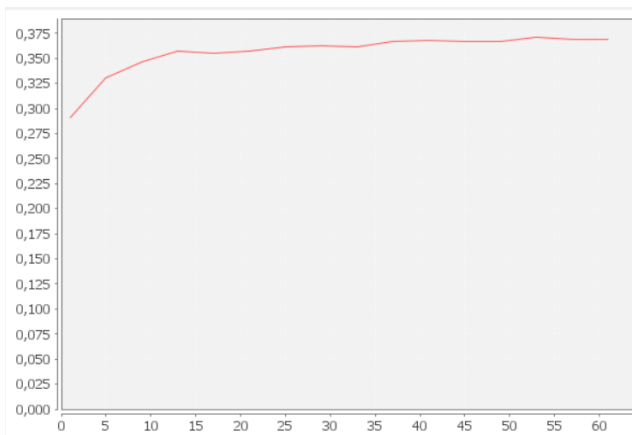


Figure 8: Evolution of k-NN accuracy as the number of neighbors increases

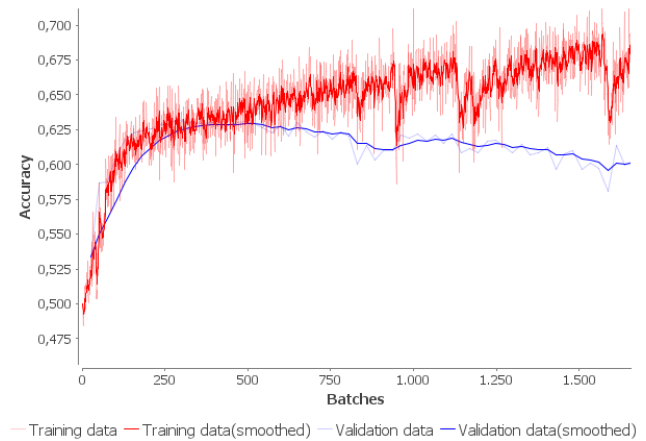


Figure 9: Learning curve of the Keras classifier

For the predictor, it has been decided to establish a fixed number of neurons, 100, and decide the best way to distribute them in different layers for each attribute using cross-validation partitioning.

On the other hand, in the neural network of Keras, The learning curve has been used to determine the number of iterations that maximizes the accuracy of cross-validation, since with a high number of iterations, the model becomes overfitted, as can be seen in Figure 9.

Finally, in the case of Random Forest, The only parameters that can be improved are the number of trees and the maximum depth of each tree, but accuracy generally increases with the number of trees, albeit exponentially decreasing. Therefore, a fixed tree size has been chosen to limit computation time.

5.3.3. Results

The results obtained with each classifier are shown in Figure 11, which shows that the relative precision of the different classification methods is essentially the same for each attribute.



Figure 10: Best relative accuracy in each attribute

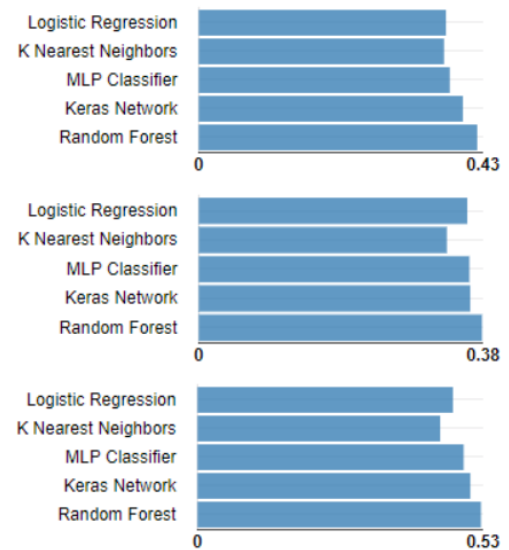


Figure 11: Accuracy by model

The most accurate model in all three cases is Random Forest, with accuracies of 42,1%, 37,9% and 52,6%. In order to accurately measure the accuracy of these classifiers, it is necessary to compare them with the trivial classifier,

as shown in Figure 10, which includes a comparison of the relative accuracies of the best model for each attribute.

As can be seen, in each of the three attributes a classifier between 2 and 3 times better than the trivial one has been obtained.

6. Conclusions

This project has highlighted the limitations imposed by the time complexity of learning and data mining algorithms, as the execution time of some of them has posed a significant obstacle for a medium-sized dataset. In this regard, the usefulness of dimensionality reduction methods such as PCA and t-SNE has been observed, both for data visualization and for reducing the complexity of the analysis, sometimes even making previously inapplicable algorithms applicable.

The importance of the appropriate selection of parameters in learning models has also been observed, particularly in classification algorithms, as well as the usefulness of different cross-validation techniques.

Regarding the problem discussed, unfortunately it has not been possible to train predictors that we can call reliable - although better than the trivial one - nor to extract results that clarify the uncertainty of our future, but in the process we have acquired some more knowledge about a panorama that is not current, but is recent.

In a way, it can even be considered comforting that our future is so difficult to predict, and does not depend solely on the few parameters collected by the survey we have studied.

References

Knime Hub. Elbow Method - hayasaka. <https://hub.knime.com/hayasaka/spaces/Public/latest/ScienceCafe-2021-01-25/ElbowMethod~Bk0qD38ZfLPn7YXQ>, 2022. [Accessed May 22, 2022].